

Course Code	21ECO550T	Course Name	Applied Machine Learning in Imaging Systems	Course Category	Open Elective	L	T	P	C
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Pre-requisite Courses	NIL	Co- requisite Courses	NIL	Progressive Courses	NIL
Course Offering Department	Electronics and Communication Engineering		Data Book / Codes / Standards	NIL	

Course Learning Rationale (CLR):	<i>The purpose of learning this course is to:</i>
CLR-1:	<i>Introduce the fundamentals of imaging systems and machine learning</i>
CLR-2:	<i>Get familiar with the concepts of image preprocessing and feature extraction</i>
CLR-3:	<i>Explain the need of machine learning models for image processing</i>
CLR-4:	<i>Acquire knowledge on various models used for object detection and localization</i>
CLR-5:	<i>Utilize the concepts of machine learning applied to imaging systems.</i>

Course Outcomes (CO):	<i>At the end of this course, learners will be able to:</i>	Programme Outcomes (PO)		
		1	2	3

CO-1:	<i>Gain knowledge on the fundamentals of digital image processing and machine learning</i>	3	-	1
CO-2:	<i>Interpret preprocessing and feature extraction methods on digital images</i>	3	-	1
CO-3:	<i>Inspect the use of machine learning models for the processing of digital images</i>	3	-	2
CO-4:	<i>Analyze the various object detection and localization models</i>	3	-	2
CO-5:	<i>Develop and implement machine learning solutions for various imaging problems</i>	-	3	3

Module-1 - Introduction to Imaging Systems and Machine Learning	9 Hours
<i>Introduction to Imaging Systems: Fundamentals of digital imaging, Image acquisition and representation, Common imaging systems and their applications. Overview of Machine Learning: Basics of machine learning, Types of machine learning algorithms and models - supervised, unsupervised, reinforcement learning.</i>	
Module-2 - Image Preprocessing and Feature Extraction	9 Hours
<i>Image Preprocessing Techniques: Noise reduction, Contrast enhancement, Image normalization and scaling. Feature Extraction: Edge detection, Texture analysis, Color and shape features. Dimensionality Reduction: Principal Component Analysis (PCA), Linear Discriminant Analysis (LDA) and t-distributed Stochastic Neighbor Embedding (t-SNE).</i>	
Module-3 - Machine Learning Models for Image Processing	9 Hours
<i>Supervised Learning for Image Classification: Decision trees, random forests, and support vector machines, K-nearest neighbors and logistic regression, Performance metrics and evaluation. Unsupervised Learning for Image Segmentation: Clustering techniques (K-means, hierarchical clustering), Gaussian Mixture Models (GMM), Evaluation of segmentation results.</i>	

Module-4 - Models for Object Detection and Localization	9 Hours
<i>Deep Learning: Introduction to neural networks, Convolutional Neural Networks (CNNs) for image recognition, Transfer learning and pre-trained models. Object Detection and Localization: YOLO (You Only Look Once), R-CNN (Region-based Convolutional Neural Networks), SSD (Single Shot MultiBox Detector). Generative Models and Image Synthesis: Generative Adversarial Networks (GANs).</i>	
Module-5 - Case Studies and Practical Applications	9 Hours
<i>Medical Imaging: Machine learning in radiology and pathology, Case studies in medical image analysis; Remote Sensing and Satellite Imaging: Applications in environmental monitoring and agriculture Techniques for processing satellite imagery; Project Development: Project proposal and planning, Implementation of machine learning solutions in imaging systems.</i>	

Learning Resources	1. Rafael C Gonzalez, Richard E Woods, "Digital Image Processing"- 4th Edition, Pearson Education 2018. 2. Simon O. Haykin, "Neural Network and Learning Machines", 3rd ed., Pearson Education, 2009. 3. Milan Sonka, Vaclav Hlavac, Roger Boyle, "Image Processing, Analysis, and Machine Vision with Mind Tap", 3rd Ed, 2008, Thomson Learning.	4. Christopher M. Bishop, "Pattern Recognition and Machine Learning" by, Springer, August 2006. 5. Jeff Proise, "Applied Machine Learning and AI for Engineers", O'Reilly Media, Inc., November 2022. 6. Research papers and case studies provided throughout the course.
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Learning Assessment				
	Bloom's	Continuous Learning Assessment (CLA)		Summative
		Formative	Life-Long Learning	Final Examination

	Level of Thinking	CLA-1 Average of unit test (50%)		CLA-2 (10%)		(40% weightage)	
		Theory	Practice	Theory	Practice	Theory	Practice
Level 1	Remember	15%	-	15%	-	15%	-
Level 2	Understand	25%	-	20%	-	25%	-
Level 3	Apply	30%	-	25%	-	30%	-
Level 4	Analyze	30%	-	25%	-	30%	-
Level 5	Evaluate	-	-	10%	-	-	-
Level 6	Create	-	-	5%	-	-	-
	Total	100 %		100 %		100 %	

Course Designers		
Experts from Industry	Experts from Higher Technical Institutions	Internal Experts
1. Mr. Kartik Soundarajan, Vice President, Quadgen Wireless Solution Pvt. Ltd., Bangalore.	1. Dr. M. Meenakshi, College of Engineering, Guindy.	1. Dr. S. Dhanalakshmi
2. Mr. George Jacob, CEO and Founding member, Semicon Design Technologies, Bangalore		0. Dr. M. Sangeetha